

Quick Check

Lesson 1: Noticing the Phenomenon & Building the Need: What Tool Is Missing?

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 2.4: Trigonometry 1 | Lesson 1

A student stands 40 m from the base of a straight transmission tower and estimates that the angle up to the top is about 50° . The class tries to find the height using only algebra and Pythagoras. Which statement best explains why they get stuck?

- A** Pythagoras needs two sides of a right-angled triangle, but here only one side (40 m) and an angle are known.
- B** Pythagoras can be used directly because 40 m and 50° are both known.
- C** The tower does not form a right-angled triangle, so no method could ever work.
- D** Pythagoras only works for triangles taller than 40 m.

A wheelchair ramp forms a right-angled triangle with the level ground and a vertical wall. The ramp is the sloping side. You stand at the angle where the ramp meets the ground. Which side is the **ADJACENT** side to that angle?

A The ramp (the sloping side)

B The level ground

C The right angle itself

D The vertical wall

In Part B, a group draws a right-angled triangle with a 40° angle. They measure the opposite side as 3.9 cm, the adjacent side as 4.6 cm and the hypotenuse as 6.0 cm. What is $O \div H$ to two decimal places?

A 0.77

B 0.85

C 0.65

D 1.54

A group draws three right-angled triangles that all have a 40° angle. Their hypotenuses are about 6 cm, 9 cm and 12 cm. They find $O \div H = 0.64$, 0.65 and 0.64. What does this result show?

- A** The $O \div H$ values are almost the same even though the triangles are different sizes.
- B** $O \div H$ is always equal to the length of the hypotenuse.
- C** $O \div H$ gets bigger as the triangle gets bigger.
- D** The group must have measured wrongly, because different triangles should give very different values.

In the mast problem, the right angle is at the base of the mast. The observer stands 40 m from the base, and the angle from the observer up to the top is 50° . The height is the side opposite the 50° angle, and the 40 m is the side adjacent to it. Which equation connects the height to the 40 m?

A $\text{height} \div 40 = \sin 50^\circ$

B $\text{height} \div 40 = \cos 50^\circ$

C $\text{height} \div 40 = \tan 50^\circ$

D $\text{height} \div 40 = \tan 40^\circ$

Which of these is the best question to post on the Driving Question Board for the mast problem?

A Is the mast in Nakuru?

B Why do we have to do trigonometry?

C What is maths?

D How can we find the value of $\tan 50^\circ$ without measuring a huge triangle?

When drawing a model of the mast scenario (a vertical mast standing on level ground, with the observer 40 m away), where should the right angle be marked?

- A** Halfway along the 40 m line
- B** At the top of the mast
- C** At the observer's feet, where the 50° angle is
- D** At the base of the mast, where the vertical mast meets the level ground

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).